Indian Statistical Institute

Mid-semsetral examination : (2025-26)

M. Math. II year

Differential Geometry I

Date : 08.9.25 Maximum marks : 30 Duration : 2 hours.

Answer ANY THREE questions. Each carries 10 marks. A surface means a level set of some regular smooth map as discussed in class.

(1) Compute the signed curvature at every point of the oriented planar curve $\{(x,y) \in \mathbb{R}^2 : f(x,y) \equiv x^2 - y^2 = 1,\ x > 0\}$, with the usual orientation given by $\frac{\nabla f}{\|\nabla f\|}$.

(2) Prove that a smooth regular curve γ in $\mathbb{R}^3$ is a planar curve (i.e. the image of γ lies in a plane) if and only if the torsion function τ is identically zero.

(3) Prove that any smooth vector field on a compact surface is complete.

(4) Let S be the cylinder in $\mathbb{R}^3$ given by $\{(x,y,z) : x^2 + y^2 = 1\}$. Compute the mean curvature and Gaussian curvature at any point of S.

$(2f, 2y, 0)$